Automatic Piano Transcription: A Comparison of Traditional Signal Processing and CNN-Based Deep Learning Models

Abstract

Automatic Music Transcription (AMT) aims to convert raw audio signals into symbolic musical representations such as MIDI. Piano transcription is particularly challenging due to the polyphonic nature of the instrument and overlapping harmonic structures. This paper presents a convolutional neural network (CNN)-based approach for automatic piano transcription using mel spectrogram representations of audio signals. The model is trained on the MAESTRO dataset and predicts a piano roll representation indicating active notes over time. Exploratory data analysis on the dataset revealed a 5% note density and average polyphony of 4.6 notes per frame, findings that directly shaped the loss function design and data loading strategy. Three approaches are compared: a traditional signal processing baseline (pYIN), a CNN + BiLSTM model, and a CNN + Transformer model. The CNN + Transformer achieves the best result with an F1 score of 0.816, demonstrating that learned polyphonic models substantially outperform monophonic baselines. Results and evaluation highlight the strengths and limitations of each approach, providing insights for future improvements.

1. Introduction

Automatic Music Transcription (AMT) is the process of converting raw audio recordings into symbolic representations such as MIDI. For piano music, this involves identifying which of the 88 keys are active at any given time, along with their onset, offset, and intensity. This task is challenging due to the polyphonic nature of piano music, where multiple notes are played simultaneously, resulting in overlapping harmonic frequencies.

Each piano note produces not only a fundamental frequency but also a series of harmonics, which makes distinguishing between notes difficult when they occur together. Additional challenges include variations in dynamics, recording conditions, and performer style. These complexities make AMT a difficult problem that lies at the intersection of signal processing, machine learning, and music theory.

Recent advances in deep learning have significantly improved performance in AMT systems. Many modern approaches rely on neural networks trained on large datasets using time-frequency representations such as mel spectrograms. These representations convert audio into image-like formats, enabling the use of convolutional neural networks (CNNs).

Prior work by Kong et al. proposes a high-resolution transcription system that regresses onset and offset times, achieving strong performance on the MAESTRO dataset. Similarly, Gardner et al. introduce the MT3 model, which uses Transformer architectures for multi-instrument transcription. While these approaches achieve state-of-the-art results, they require substantial computational resources.

In this work, we propose and compare three approaches for piano-only transcription: a traditional signal processing baseline, a CNN + BiLSTM model, and a CNN + Transformer model, all trained from scratch on the MAESTRO dataset. The goal is to evaluate the contribution of temporal

context modeling sequential versus global to frame-level transcription accuracy, while maintaining computational efficiency relative to state-of-the-art systems.

2. Methods

The proposed system implements an end-to-end deep learning pipeline for automatic piano transcription. The workflow begins by converting raw audio signals into mel spectrograms, which serve as the primary input to the neural network. A mel spectrogram represents the distribution of energy across different frequency bands over time, effectively transforming the audio signal into a two-dimensional representation where patterns can be learned spatially.

2.1 Data Representation

The input audio signals are converted into mel spectrograms, which represent the distribution of energy across frequency bands over time. This transformation converts the audio signal into a two-dimensional representation, where the horizontal axis corresponds to time, the vertical axis represents frequency, and the intensity indicates energy.

Mel spectrograms are particularly suitable for music transcription tasks because they align with human auditory perception and preserve important harmonic structures. This allows the problem to be treated as an image recognition task. Parameters were chosen to match the piano frequency range: sample rate of 22,050 Hz, n_fft = 2048, hop_length = 512, n_mels = 128, and fmax = 8,000 Hz.

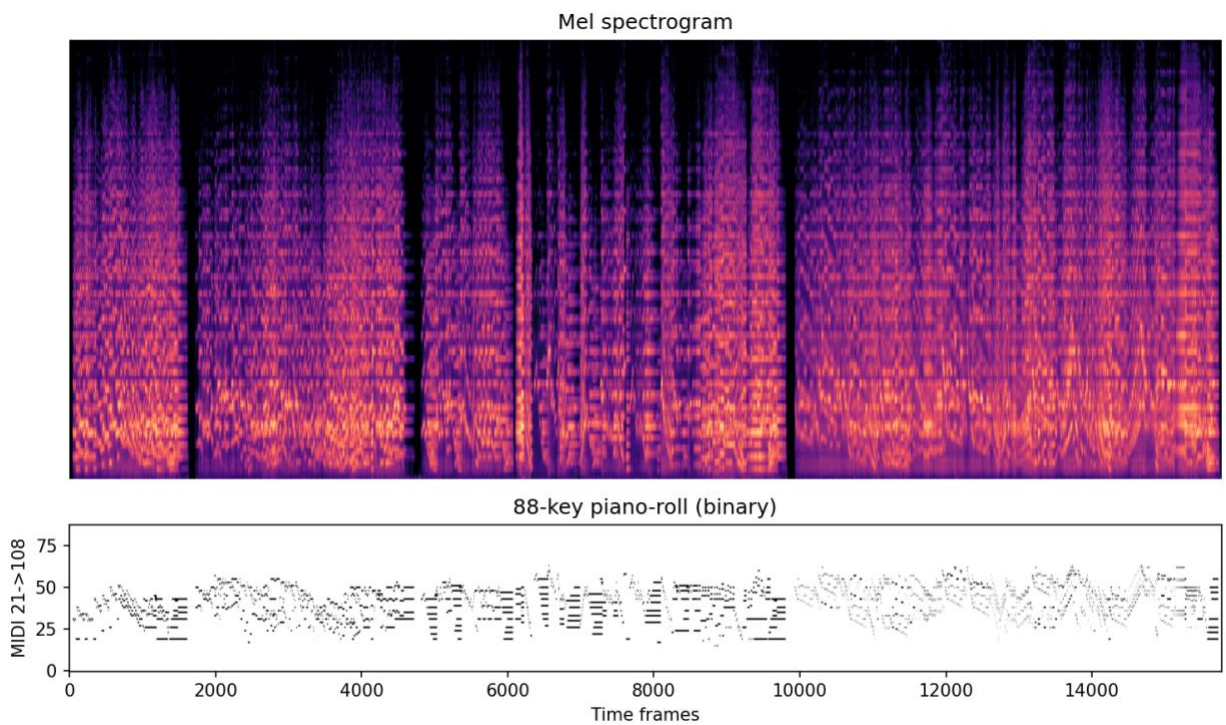

Figure 1: Example paired mel spectrogram (top) and ground-truth piano roll (bottom) for one 6-second chunk. The vertical axis of the piano roll corresponds to the 88 piano keys (MIDI 21–108).

2.2 Model Architecture

The proposed system compares three model architectures, all sharing a common CNN encoder designed to extract hierarchical features from the mel spectrogram input. The CNN encoder consists of two Conv1d layers with batch normalization and dropout, which progressively learn spectral features from basic frequency peaks to harmonic relationships between notes.

The initial layers detect basic spectral features such as frequency peaks, while deeper layers learn higher-level representations corresponding to specific piano notes. The final layer of each model outputs a piano roll representation, a matrix indicating which of the 88 piano keys are active at each time step.

Three variants are evaluated: (1) a traditional signal processing baseline with no learned parameters, (2) a CNN encoder paired with a Bidirectional LSTM for sequential temporal modeling, and (3) a CNN encoder paired with a Transformer encoder for global attention-based temporal modeling. All learned models are trained using binary cross-entropy loss with positive-class upweighting (pos_weight = 10) to address the severe class imbalance caused by a note density of only 5% (95% of frame-key pairs are silent).

2.3 Dataset

The model is trained on the MAESTRO v3 dataset, which contains approximately 200 hours of high-quality piano recordings paired with aligned MIDI annotations. The 2015 subset was used, providing 129 paired audio/MIDI recordings. This dataset provides precise ground truth labels and is widely used in AMT research.

All 129 files were preprocessed into 8,623 fixed-length chunks of 256 frames each (approximately 6 seconds per chunk). The dataset was split at the file level, 103 files for training and 26 for validation to prevent the same song from appearing in both sets (data leakage prevention).

2.3.1 Exploratory Data Analysis

Prior to model development, exploratory data analysis was conducted on the full 129-file dataset to characterize the distribution of musical content and inform preprocessing and training decisions.

Note density and class imbalance. Across all 2.2 million annotated frame-key pairs, only 5.06% are active (note-on). This means 95% of the training signal is silence, creating a severe class imbalance. Left unaddressed, a model trained with standard binary cross-entropy would achieve high accuracy by predicting all zeros. This finding directly motivated the use of positive-class upweighting (pos_weight = 10) in the loss function.

Polyphony. The average number of simultaneously active notes per frame is 4.6, with peaks reaching 10 or more in complex passages. This confirms that piano transcription is an inherently multi-label problem and rules out monophonic pitch detection methods as viable baselines for achieving high recall.

Pitch distribution. The pitch activation histogram shows that the middle register (MIDI pitches 48–84, approximately C3 to C6) accounts for the majority of note activity, while the extreme low and high registers are used sparingly. This uneven pitch distribution suggests that per-register performance may vary, which is a direction for future error analysis.

Chunk size selection. At the preprocessing parameters used (sr = 22,050 Hz, hop_length = 512), each frame corresponds to approximately 23 ms of audio. A chunk size of 256 frames therefore covers approximately 5.9 seconds — long enough to capture full note events including attack, sustain, and release phases, while remaining short enough for efficient batched training.

The 129 files yielded 8,623 non-overlapping chunks in total, providing a 67× increase in effective training examples compared to file-level batching.

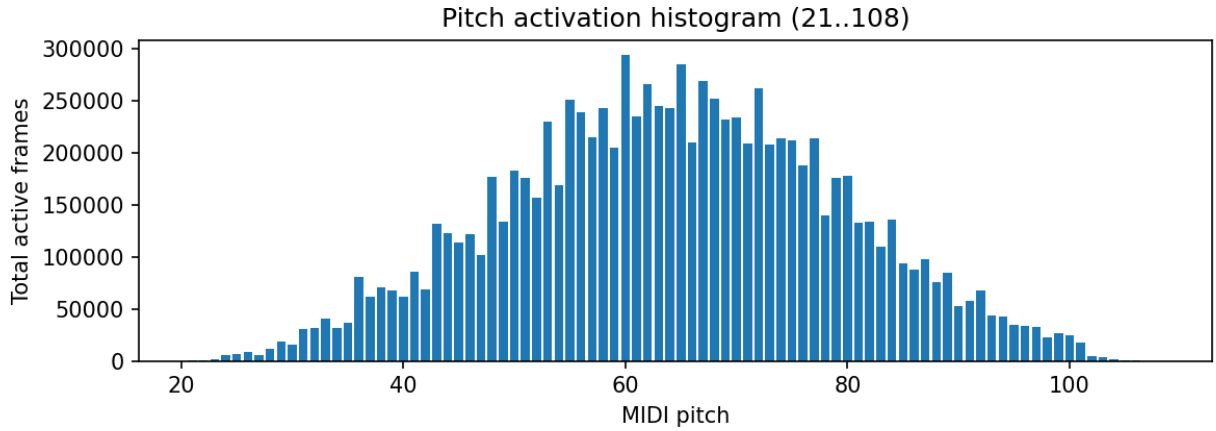

Figure 2: Pitch activation histogram across all 129 training files. The middle register (MIDI 48–84) accounts for the majority of note activity.

2.4 Training Procedure

The training process involves optimizing the model parameters using the Adam optimizer with a ReduceLROnPlateau learning rate scheduler (patience = 3, factor = 0.5). Gradient clipping (max_norm = 1.0) is applied to stabilize training. Best-model checkpointing is based on validation F1 rather than loss, since F1 directly captures the precision-recall balance under class imbalance.

No data augmentation was applied in this work. The models are trained directly on pre-computed mel spectrograms and piano roll arrays loaded from disk. Incorporating augmentation techniques such as pitch shifting and time stretching is noted as a direction for future work.

2.5 Evaluation Metrics

Model performance is evaluated using frame-level Precision, Recall, and F1 score computed across all 88 piano keys. These metrics assess how well the predicted notes match the ground truth annotations in terms of correctness and completeness. Mean and standard deviation of F1 are reported across all 26 validation files.

3. Results

The proposed system was evaluated on 26 held-out validation songs from the MAESTRO 2015 subset, which were excluded from training at the file level to prevent data leakage. Three approaches were compared: a traditional signal processing baseline, a CNN + BiLSTM model, and a CNN + Transformer model.

3.1 Traditional Signal Processing Baseline

The pYIN-based baseline achieved a Precision of 0.563 but a near-zero Recall of 0.027, yielding an F1 score of 0.050 ± 0.040 (median: 0.035). The high precision indicates that detected pitches are usually correct; however, because pYIN is fundamentally monophonic, it can only identify one pitch per frame. Given that the dataset averages 4.6 simultaneous notes per frame, roughly 97% of active notes go undetected.

3.2 CNN + BiLSTM

The CNN + BiLSTM model (3.1M parameters) consists of a CNN encoder with two Conv1d layers with BatchNorm and dropout, feeding into a 2-layer Bidirectional LSTM with 256 hidden units per direction. The BiLSTM propagates hidden state across the entire chunk, allowing it to model the full temporal lifecycle of a note (attack, sustain, release). Bidirectionality gives each frame access to both past and future context.

This model achieved Precision = 0.691, Recall = 0.902, and F1 = 0.781 ± 0.039 (median: 0.786). The strong recall demonstrates that the learned model successfully detects polyphonic note activity that is invisible to monophonic methods.

3.3 CNN + Transformer

The CNN + Transformer model (2.0M parameters) uses the same CNN encoder feeding into a 2-layer Transformer encoder with 8-head self-attention, 1024-dim feedforward layers, and sinusoidal positional encoding. Self-attention directly connects any two frames in a single operation, learning long-range harmonic dependencies. The model was trained for 50 total epochs including fine-tuning at lr = 0.0005.

This model achieved the best overall performance: Precision = 0.750, Recall = 0.899, and F1 = 0.816 ± 0.037 (median: 0.822). The improvement in precision over the BiLSTM (+0.059) suggests that global self-attention is better able to suppress false positives by directly modeling harmonic relationships across the full 256-frame context window.

3.4 Comparison Table

Model	Parameters	Precision	Recall	F1 (mean ± std)	F1 (median)
Traditional SP (pYIN)	0	0.563	0.027	0.050 ± 0.040	0.035
CNN + BiLSTM	3.1M	0.691	0.902	0.781 ± 0.039	0.786
CNN + Transformer	2.0M	0.750	0.899	0.816 ± 0.037	0.822

Below are the output of 3 different models. All four piano rolls display the same 30-second excerpt (seconds 30–60) from a single validation recording in the MAESTRO 2015 dataset, held out from training. The excerpt was selected for visual comparison purposes only; quantitative results reported in Table 1 are computed over the full 26-file validation set.

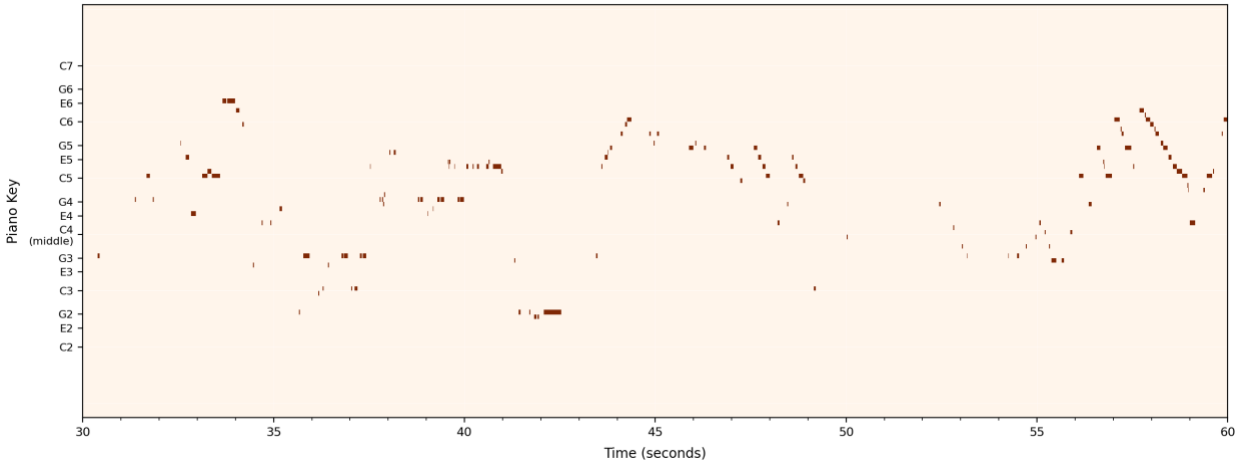

Figure 3 : Traditional Signal Processing Baseline (pYIN)

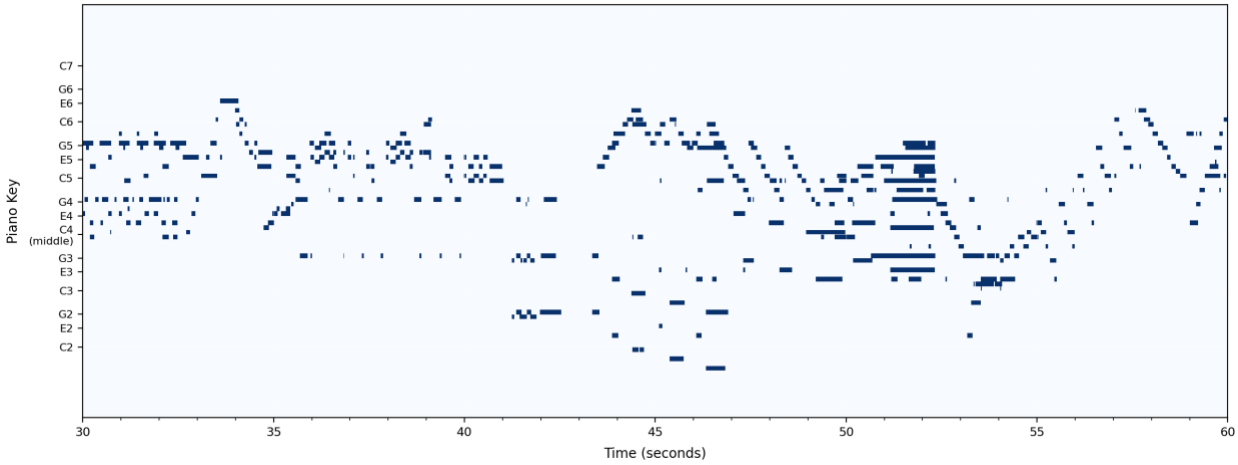

Figure 4 : CNN + BiLSTM Model

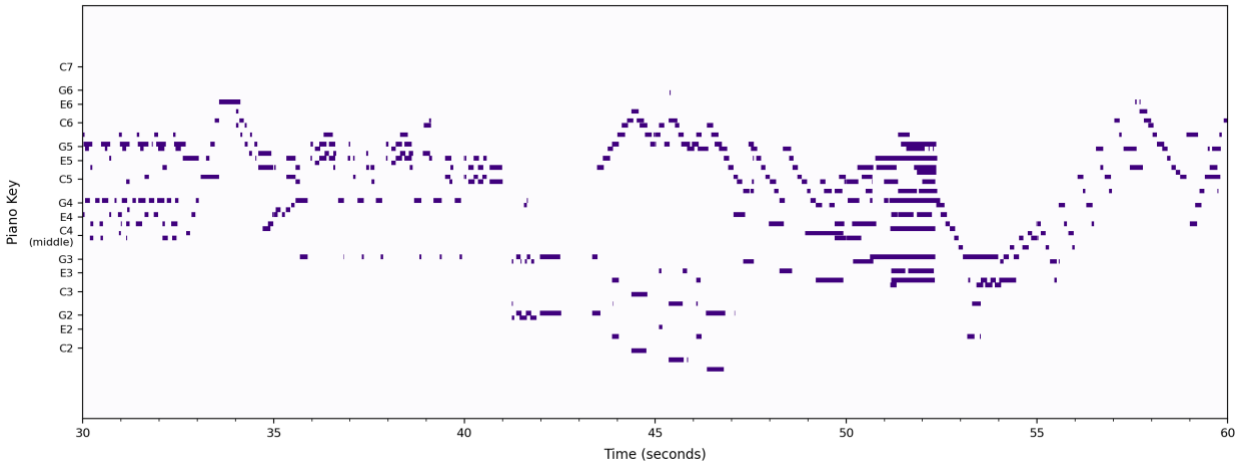

Figure 5 : CNN + Transformer Model

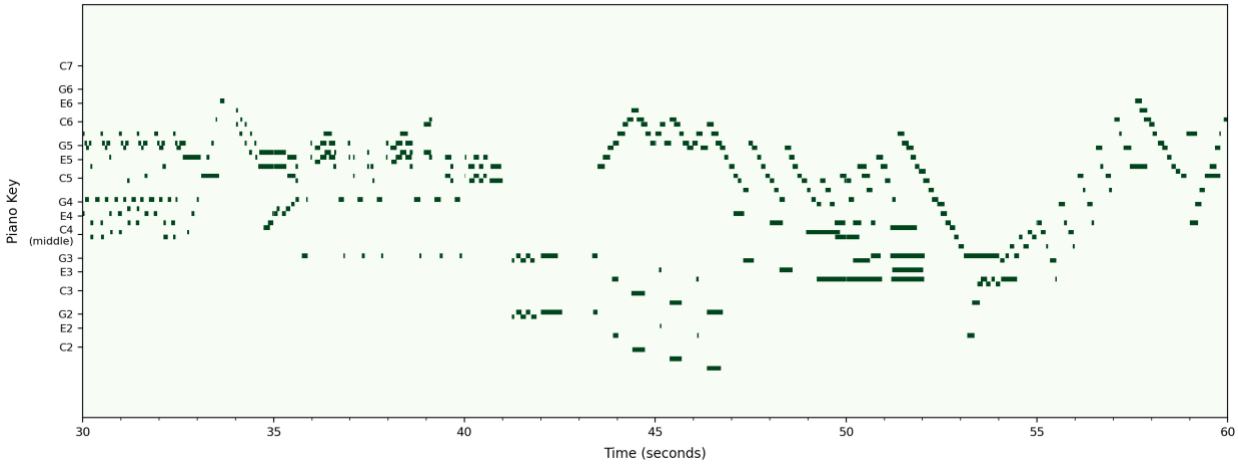

Figure 6 : Ground Truth (MAESTRO Dataset)

3.5 Observations

The 15x F1 improvement from the traditional baseline to the CNN-based models (0.050 to 0.781) demonstrates the core limitation of monophonic pitch detection for polyphonic music. The pYIN baseline can detect a single dominant pitch with reasonable accuracy, but piano music requires simultaneous detection of multiple pitches, a capability that only learned models provide.

The additional gain from BiLSTM to Transformer (+0.035 F1) reflects the advantage of global attention over sequential hidden-state propagation for capturing long-range harmonic context. The Transformer's higher precision (0.750 vs. 0.691) indicates it produces fewer false positives, likely because self-attention can directly model harmonic relationships across the full 256-frame context window rather than propagating information through hundreds of LSTM gate operations.

The CNN + Transformer model was still improving at epoch 50 (F1 = 0.816), suggesting that further training could yield additional gains. The class imbalance inherent in the dataset (95% of frame-key pairs are silent) was addressed via positive-class upweighting (pos_weight = 10), which was critical for achieving high recall without sacrificing precision.

4. Conclusion

This paper compared three approaches to automatic piano transcription: a traditional signal processing baseline (pYIN), a CNN + BiLSTM model, and a CNN + Transformer model. All three were evaluated on the same 26 held-out validation files from the MAESTRO 2015 subset, using frame-level Precision, Recall, and F1 as metrics.

The results show a clear progression. The traditional pYIN baseline achieves F1 = 0.050, demonstrating that monophonic pitch detection is fundamentally insufficient for polyphonic piano transcription. Its near-zero recall (0.027) reflects an architectural limitation, pYIN can only track one pitch at a time, so it misses roughly 97% of notes in music that averages 4.6 simultaneous pitches per frame. This limitation cannot be overcome through parameter tuning; it is inherent to the algorithm.

The CNN + BiLSTM model (F1 = 0.781) overcomes this by learning to detect simultaneous note activations from the full mel spectrogram. The CNN + Transformer model achieves the best performance (F1 = 0.816) by using global self-attention to model harmonic relationships across the full 256-frame context window, rather than propagating information sequentially through LSTM

gate operations. A key finding is that the Transformer's advantage over the BiLSTM manifests primarily in precision (0.750 vs. 0.691), suggesting that global attention is more effective at suppressing false positives in dense harmonic textures.

Limitations of the current work include the absence of data augmentation, no explicit modeling of note onset and offset times, and evaluation restricted to the 2015 subset of MAESTRO. Future directions include training on the full MAESTRO dataset, adding velocity prediction, continued training of the Transformer model beyond epoch 50 (it was still improving at the time of evaluation), and incorporating onset and offset regression as done by Kong et al. [1].

5. Works Cited

[1] Q. Kong, B. Li, X. Song, Y. Wan, and Y. Wang, "High-Resolution Piano Transcription with Pedals by Regressing Onset and Offset Times," IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 29, pp. 3707-3717, 2021.

[2] J. Gardner, I. Simon, E. Manilow, C. Hawthorne, and J. Engel, "MT3: Multi-Task Multitrack Music Transcription," in Proc. International Conference on Learning Representations (ICLR), 2022.

[3] C. Hawthorne et al., "Enabling Factorized Piano Music Modeling and Generation with the MAESTRO Dataset," International Conference on Learning Representations (ICLR), 2019.